# SEPARABILITY GOES UP IN A TOWER OF ALGEBRAIC EXTENSIONS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: The goes-up half of separability theory; the companion goes-down statement needs separability of E/F and lives in the same chapter section.

*Editorial note.* The label lemma-separable-goes-up names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Chapter lemma-irreducible-polynomials (used by the author without further mention): an irreducible polynomial over a field is separable exactly when its derivative is nonzero, and an irreducible polynomial dividing two polynomials divides their gcd
- The minimal polynomial of alpha over E divides the minimal polynomial of alpha over F in E[x] (both vanish at alpha and the F-minimal polynomial lies in E[x])

*Source reference.* Original reference 09H0: sources/stacks-fields/pinned-original-source.tex, source lines 1348–1348. Prerequisite pointer invoked at 1414; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Let $K/E/F$ be a tower of algebraic field extensions.*

(1) *If $\alpha \in K$ is separable over $F$, then $\alpha$ is separable over $E$.*
(2) *if $K$ is separable over $F$, then $K$ is separable over $E$.*

*Proof.* We will use Lemma 09H0 without further mention. Let $P$ be the minimal polynomial of $\alpha$ over $F$. Let $Q$ be the minimal polynomial of $\alpha$ over $E$. Then $Q$ divides $P$ in the polynomial ring $E[x]$, say $P = QR$. Then $P' = Q'R + QR'$. Thus if $Q' = 0$, then $Q$ divides $P$ and $P'$ hence $P' = 0$ by the lemma. This proves (1). Part (2) follows immediately from (1) and the definitions. □

*Date*: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.